SSH Key:

An SSH key is a secure way to log in to a remote server (like GitHub, VPS, or a cloud server) without using a password.

It works using public-key cryptography and comes as a pair:

- Private Key → stays on your computer (secret, never share it)
- Public Key → uploaded to the server (GitHub, server, etc.)

How work:

- The server checks your public key.
- Your computer proves it owns the matching private key.
- If they match → access granted

How to Generate an SSH Key?

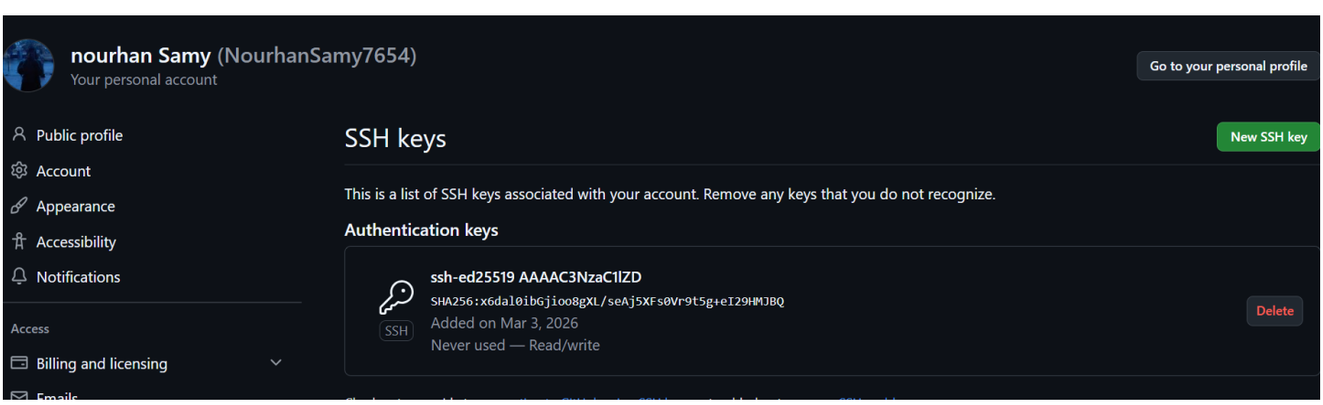
ssh-keygen -t ed25519 -C "nsamy7834@gmail.com"

- -t ed25519 → type of key
- - C → comment (usually your email)

How to Use the SSH Key?

1. Start SSH agent
   eval "$(ssh-agent -s)"
2. Add the key
   ssh-add ~/.ssh/id_ed25519
3. Copy public key

cat ~/.ssh/id_ed25519.pub

Use in my device :

```
MINGW64:/c/Users/Tech Shop
Tech Shop@DESKTOP-A42IS2E MINGW64 ~
$ ssh-keygen -t ed25519 -C "nsamy7834@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/Tech Shop/.ssh/id_ed25519):
/c/Users/Tech Shop/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase for "/c/Users/Tech Shop/.ssh/id_ed25519" (empty for no passphra
se):
Enter same passphrase again:
Your identification has been saved in /c/Users/Tech Shop/.ssh/id_ed25519
Your public key has been saved in /c/Users/Tech Shop/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:x6dal0ibGjioo8gXL/seAj5XFs0Vr9t5g+eI29HMJBQ nsamy7834@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|         o.E     |
|      o . . .    |
|     . o   o     |
|      .  .o      |
| .   o  S.+...   |
|. ..o. . oo*O.   |
| o o+.o ..*=oB   |
|o ++.... =o.* .  |
|oooo=o  oo.o .   |
+----[SHA256]-----+

Tech Shop@DESKTOP-A42IS2E MINGW64 ~
$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIJdKXANaEBYJF0fhftFB6b3FoagjV4+eM/G8UzlueYUa
 nsamy7834@gmail.com


Tech Shop@DESKTOP-A42IS2E MINGW64 ~
$

Tech Shop@DESKTOP-A42IS2E MINGW64 ~
$ eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519
bash: Shop/.ssh/agent/s.8VZRM4WKQv.agent.ns89Z7DjbF: No such file or directory
Agent pid 1840
Could not open a connection to your authentication agent.

Tech Shop@DESKTOP-A42IS2E MINGW64 ~
$ ssh -T git@github.com
The authenticity of host 'github.com (140.82.121.3)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Enter passphrase for key '/c/Users/Tech Shop/.ssh/id_ed25519':
Hi NourhanSamy7654! You've successfully authenticated, but GitHub does not provide shell access.

Tech Shop@DESKTOP-A42IS2E MINGW64 ~
$
```